Navigating Emerging Trends for the Economic Development of Second-life EV Batteries

Conrad Nichols – Senior Technology Analyst, IDTechEx

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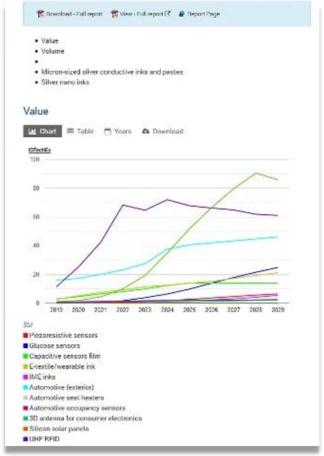

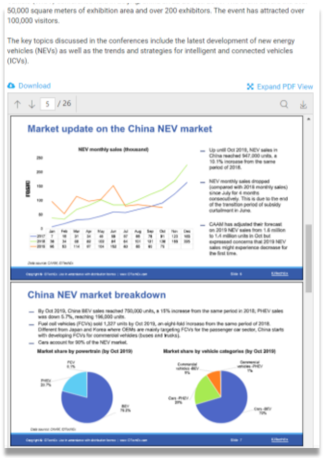

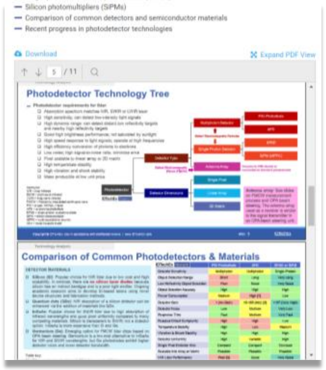

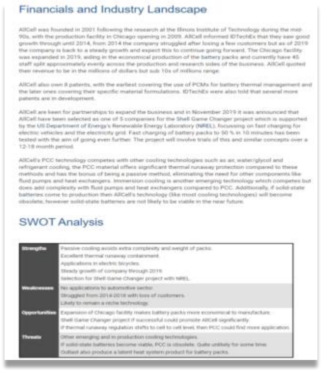

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Second-life Electric Vehicle Batteries 2025-2035: Markets, Forecasts, Players, and Technologies

10-year forecasts: second-life EV batteries & retired EV battery availability. Analyses: second-life battery technologies, markets, repurposer & end-of-life diagnostics players, regulations, repurposing costs & techno-economic analysis vs first-life Li-ion BESS.

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Webinar Agenda:

- Second-life EV Battery Market, Technology and Player Overview with Introduction to Policies / Regulations
- Cost Bottlenecks in the Repurposing Process
- Trends to Impact Repurposing Costs
- IDTechEx outlook

Second-life EV Battery Market, Technology and Player Overview

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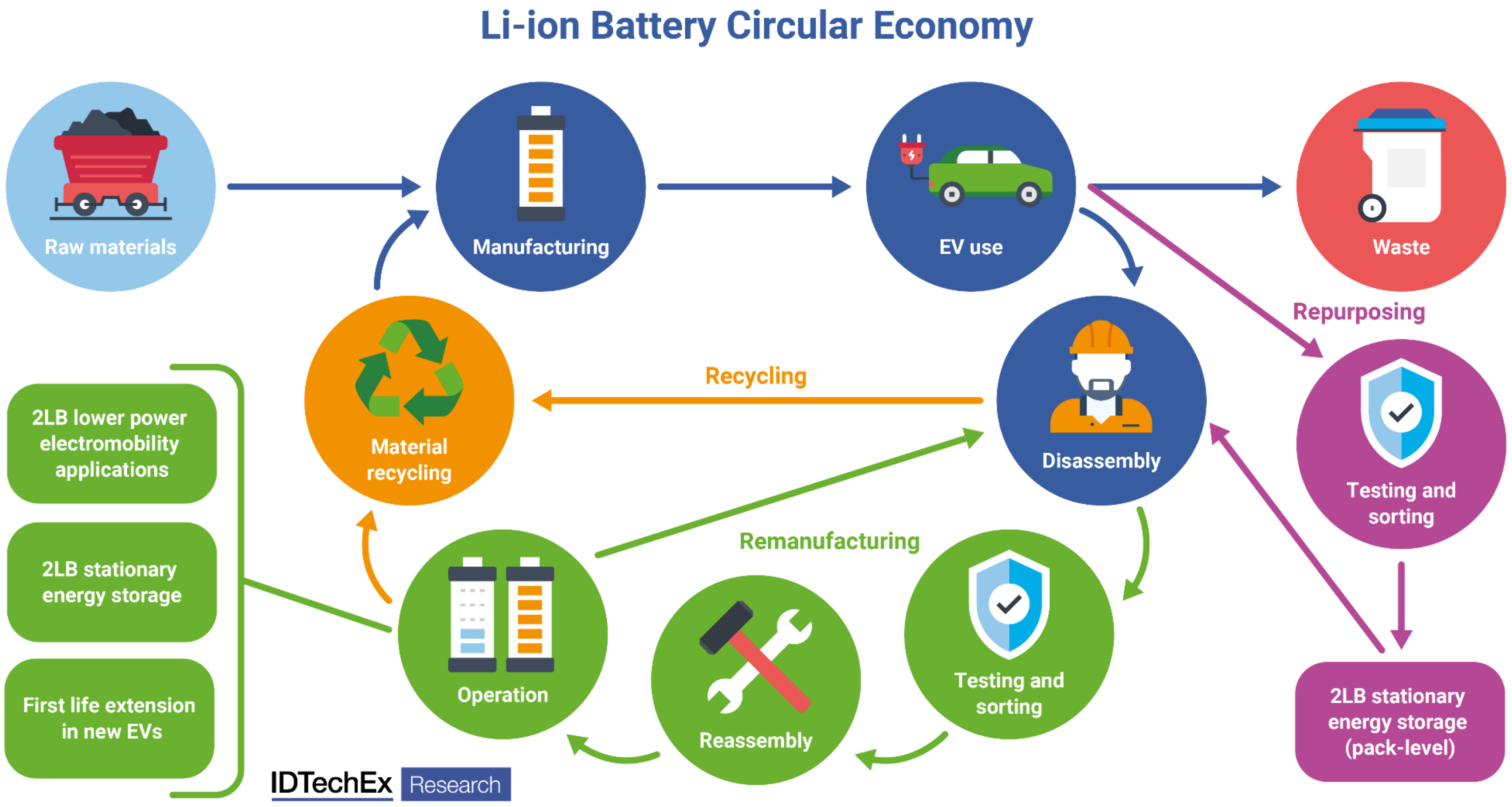
Introduction to Li-ion Battery Circular Economy



Note: 2LB = second life battery

Key Players in the Battery Second-life Value Chain

e.g. players

Battery collector

Batteries are retired from EVs and collected by EV/battery manufacturers.

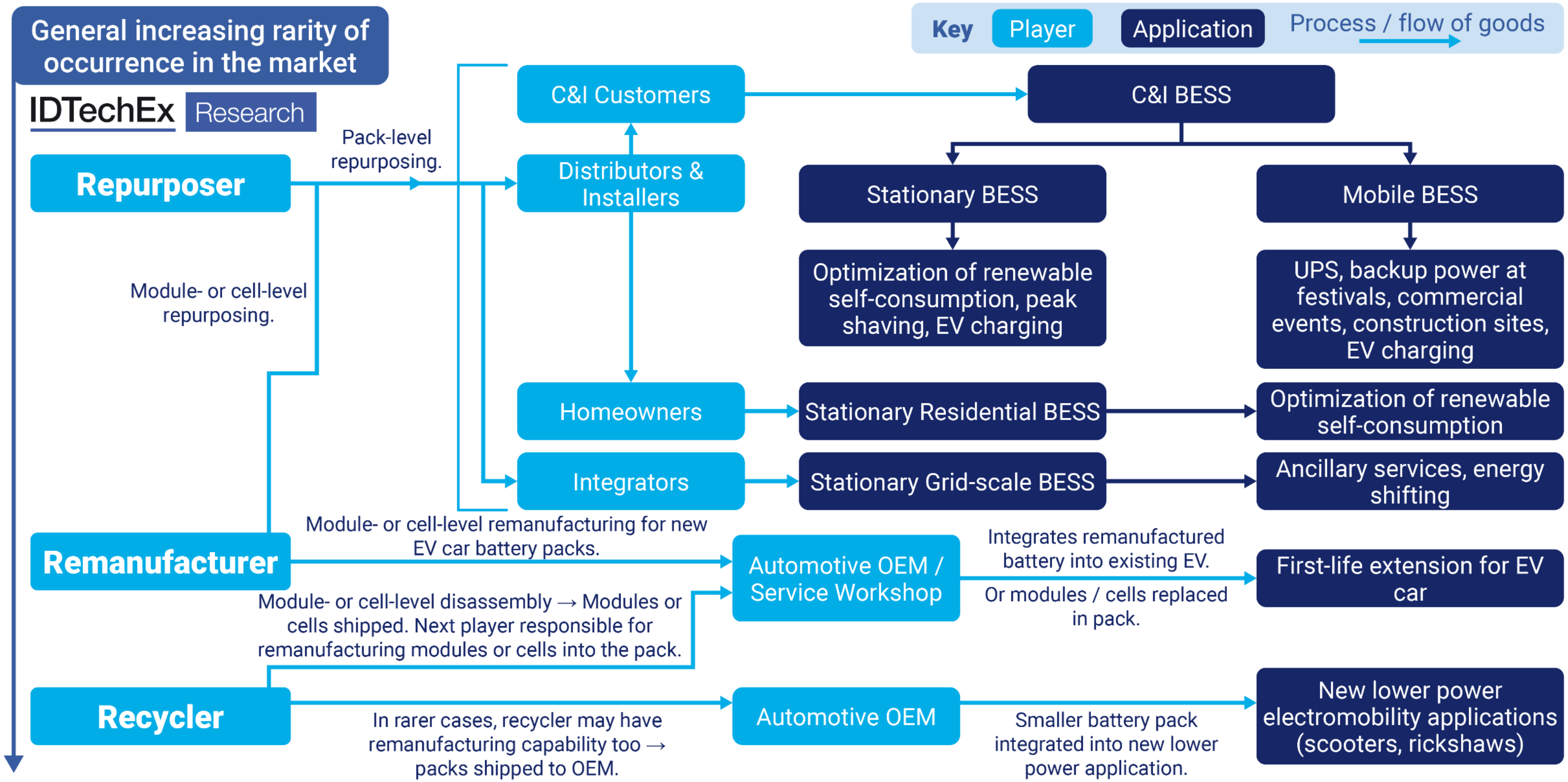
Battery performance modeler/tester

Battery performance is tested/graded or modeled from procured retired EV batteries.

Battery remanufacturer / repurposer

Batteries are tested, then repurposed or remanufactured for second use.

Second-life Battery Applications and Supply Chain Overview



Second-life Repurposers and Remanufacturers by HQ

European Players

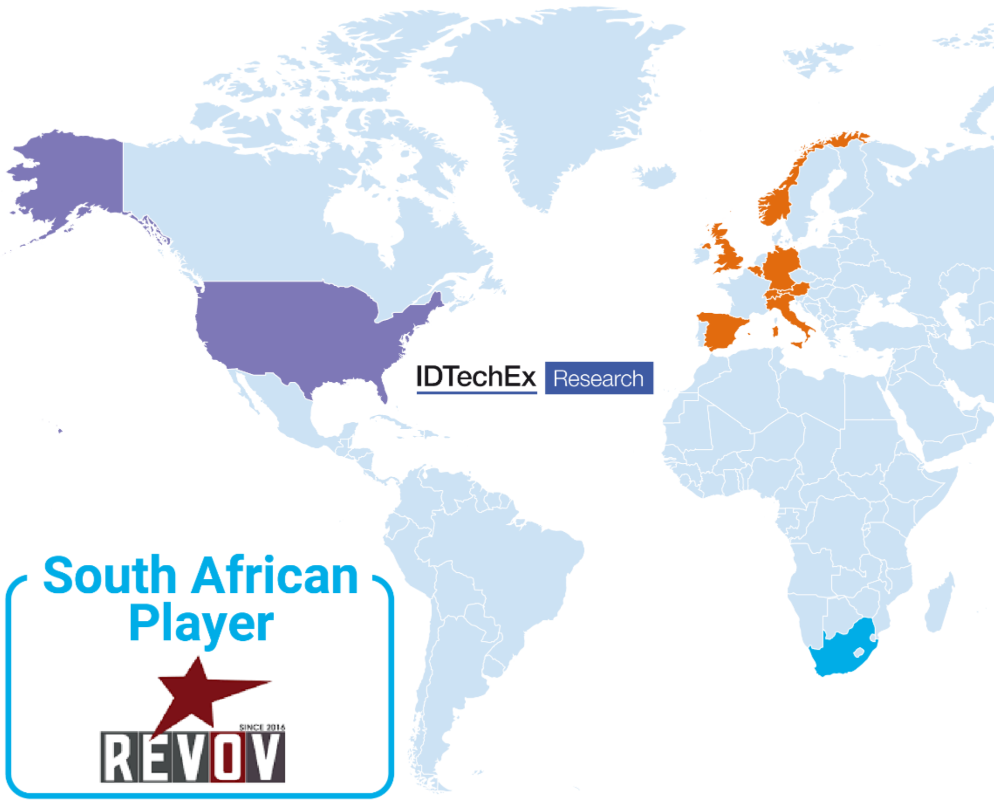
ALLYE, ALTABATT, BeePlanet factory, betteries, Connected Energy, covalion, ebs, ecostor, fenecon, libattion, Little Energy, modual, octave .energy, reefilla, ReVolta, STABL, upvolt, Voltfang, WATT4EVER, ZENOBĒ

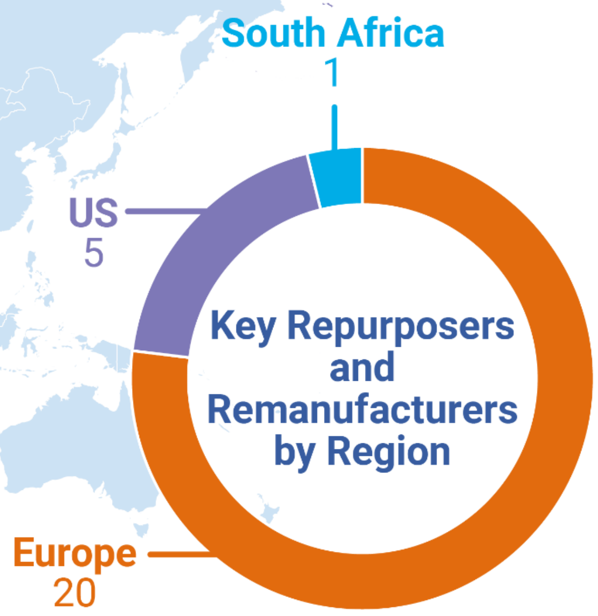
US Players

B2U, higherwire, Repurpose Energy, smartville, Terrepower

South African Player

REVOV





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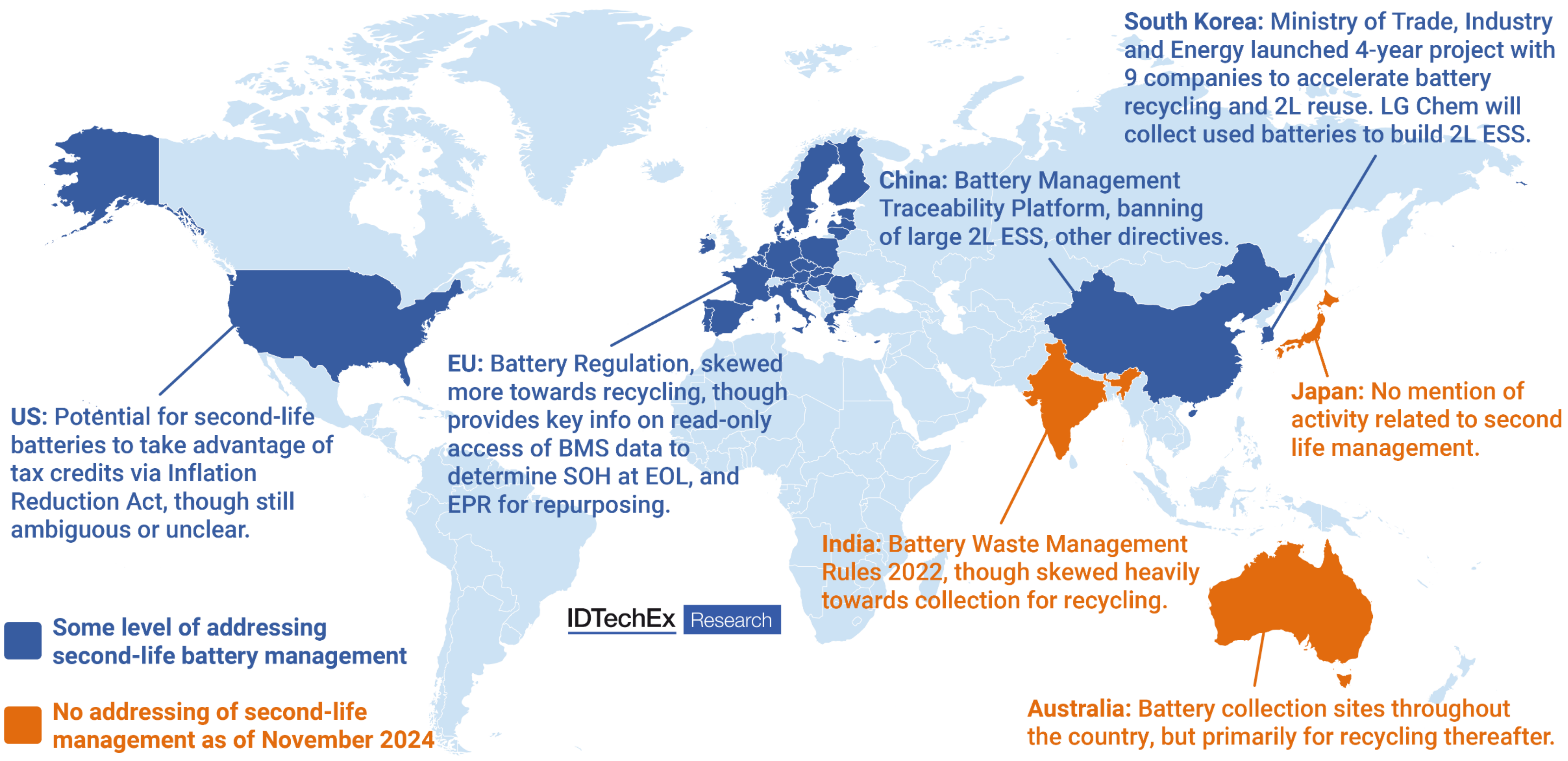
Global Second-life EV Battery Regulatory Landscape



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Influence of EU Battery Regulation

Target type	Targets	
Collection rate (LMT batteries)	**By 2029**	**By 2032**
	51%	61%
Collection rate (portable batteries)	**By 2028**	**By 2031**
	63%	73%
Overall recycling efficiency (all Li-ion batteries)	**By 2026**	**By 2031**
	65%	70%
Specific materials recovery efficiency	**By 2028**	**By 2032**
	90% cobalt 90% nickel 90% copper 50% lithium	95% cobalt 95% nickel 95% copper 80% lithium
Minimum recycled contents in new industrial and EV batteries	**From 18 August 2031**	**From 18 August 2036**
	16% cobalt 6% lithium 6% nickel 85% lead	26% cobalt 12% lithium 15% nickel 85% lead

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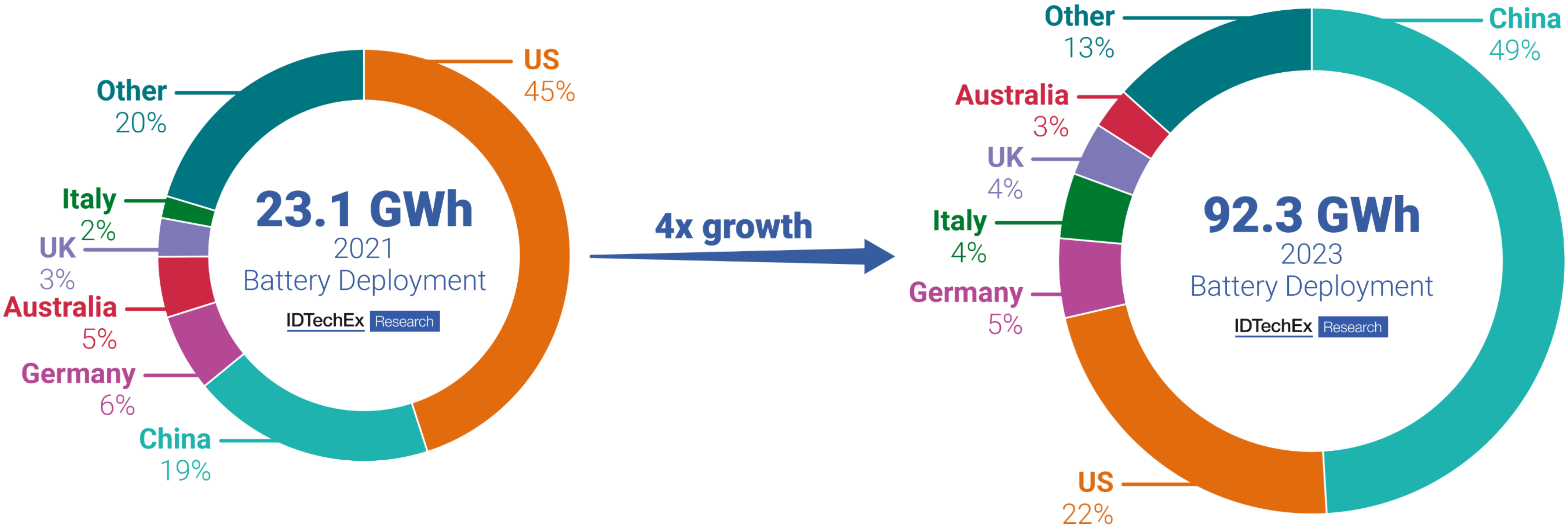
Trends to Impact Costs of Second-life Battery Repurposing

State of the First-life Li-ion BESS Market

Global deployments of first-life Li-ion BESS made annually have grown 4x in 2023 vs 2021. Some Li-ion BESS manufacturers are already reaching annual GWh-scale deployments. Prices of first-life Li-ion BESS fall into the range US$210-350/kWh in Europe & US.



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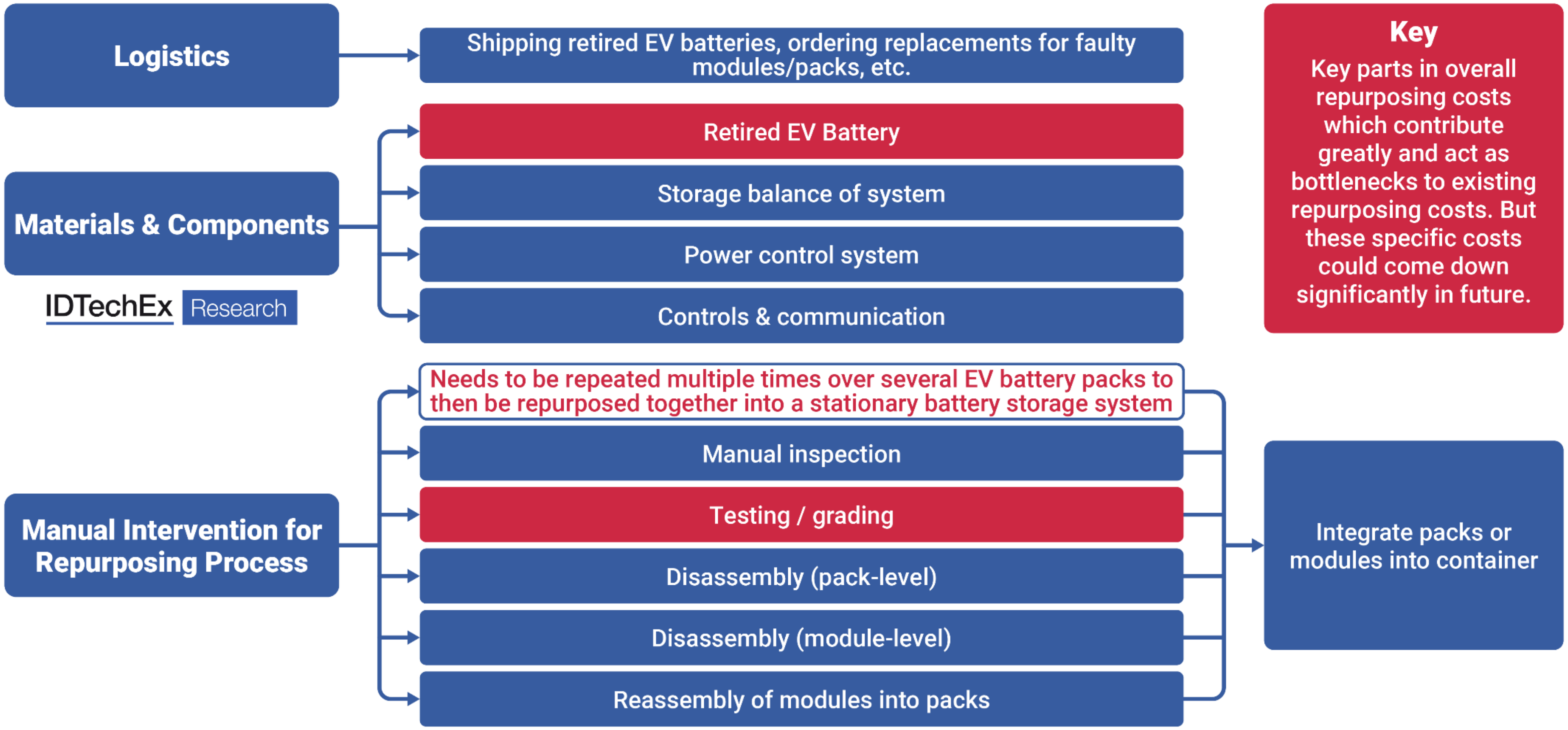
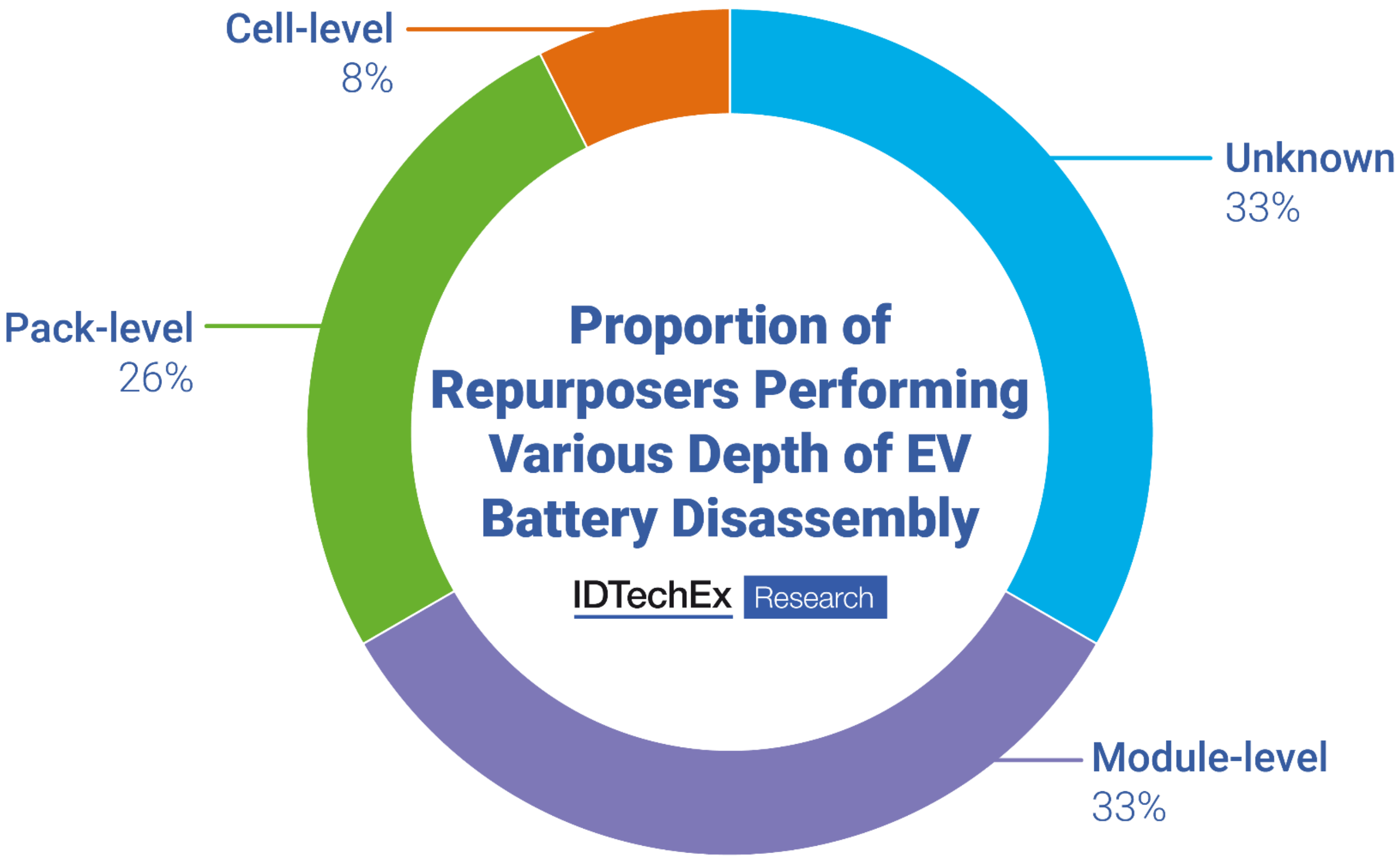
Second-life BESS Cost Comparison to First-Life Li-ion BESS and Repurposing Process Cost Bottlenecks

Second-life battery repurposing processes amount to ~US$250/kWh for one battery energy storage system (BESS). Final second-life BESS prices are therefore higher than this, and repurposers may be struggling to compete on final system price against new, first-life Li-ion BESS (US$210-350/kWh). This has led to only steady second-life EV battery market growth in 2023 and 2024.



Second-life Battery Repurposers by Depth of Disassembly

Of those repurposing processes identified, module-level and pack-level repurposers dominate the market. Pack-level repurposing offers the cheapest repurposing process, though module-level repurposing could offer improved system performance.



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Emerging B2B Marketplaces to Benefit Second-life Battery Market

B2B Marketplaces for the trading of:
- Retired EV batteries
- Second-life batteries

and potentially
- First-life batteries
- Black mass (Circunomics)

from automotive OEMs or other battery players

Marketplace or platform providers

CURRENTS MARKETPLACE

circunomics

cling + volytica diagnostics

Key information

Offers purchase prices of retired EV batteries from US$25-50/kWh. January 2024: Inventory of 2 MWh of retired EV batteries from Nissan.

June 2024: Has listed 2.5 GWh batteries across first-life and retired batteries. 2023: Traded 280 MWh combined across these two battery types.

January 2023 – 400 MWh retired EV batteries available on their platform, though not all traded. Partnership with volytica diagnostics, which monitor battery data and can provide SOH. volytica / Cling Systems' revenue generation mechanisms / business model is proprietary as of October 2024.

Key benefit: transparency of battery information (including availability and SOH) → faster trading of second-life batteries → market growth

These batteries / black mass are then supplied to:
- Second-life repurposers
- Recyclers
- Other battery players (modelers?)

Marketplace players may take a % of commission of products traded, with most going back to the automotive OEM.

Marketplace may have access to BMS data, tracking SOH over time of batteries in the field → and then displayed on marketplace for each battery once retired.

Automotive OEMs / battery players may pay for access to the marketplace or platform, and/or receive access in exchange for sharing battery data – though this type of business model has not been confirmed.

smartville **Smartville Whole Battery Catalog™ - Repository of Battery Data**

This is likely less-so of a "B2B marketplace". But generally, will still help to facilitate the tracking of data, and will allow different stakeholders to communicate and share data, likely privately. This could then still help to expedite larger volumes of retired EV battery shipments to repurposers.

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Advanced Battery Testing Technologies

Existing Situation

Battery cycling → to determine battery capacity and State-of-Health (SOH).

+

Electrochemical impedance spectroscopy (EIS) → to determine internal resistance.

→

2 – 30 hours per EV battery pack → high overhead costs. Cyclers also can exhibit high CapEx.

Future Technologies and Advantages

Advanced battery grading technologies → machine-learning assisted.

→

To determine key metrics for second-life suitability, e.g., SOH, internal resistance.

→

Order of minutes per EV battery pack with high accuracy → faster times and lower overhead costs.

↓

Could test batteries while still in vehicle via OBD port of BMS or EV charging port.

→

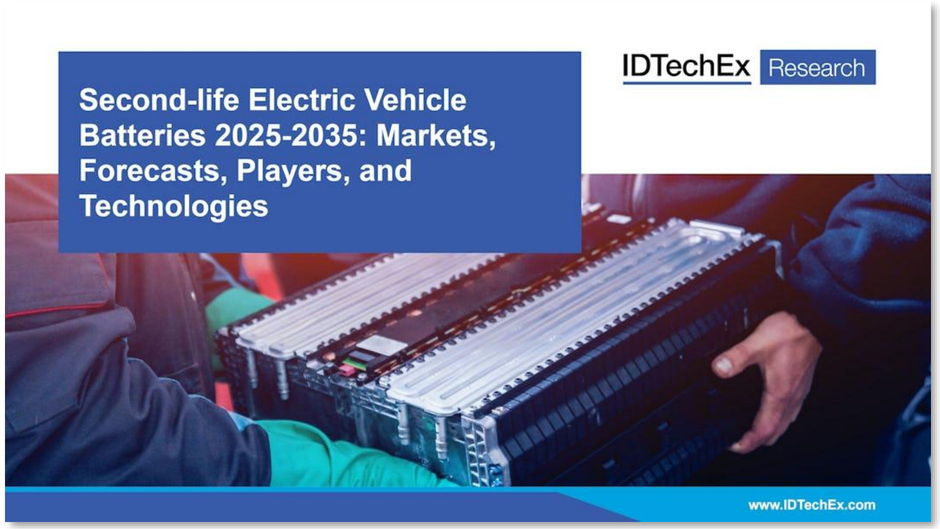
Used at service vehicle workshops & dealerships → streamlines grading process for OEMs.

→

Could reduce onus of grading from the repurposer → grading costs shared between parties?

Second-life Battery Testing and Assessment Player Summary

Player	Technology	Location	No. employees	Speed / accuracy	Notes – acquisitions, partnerships, market activity, etc.
ReJoule	EIS at pack level, in-vehicle possible. Hybrid modelling.	California, US	15	SOH: 95% accuracy / 10 mins per pack.	Per-battery modelling, looking to expand to general chemistries.
volytica diagnostics	EIS at pack level, in-vehicle possible. Data-focused.	Dresden, Germany	30	N/A	Primarily BMS diagnostics. Partnered with Cling Systems for second-life.
+ Other key second-life battery testing and assessment players in IDTechEx's Second-life Electric Vehicle Batteries 2025 – 2035 market report.					



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Semi-automated EV Battery Disassembly

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Benefit	To reduce reliance on manual labor, reducing time and costs for disassembly.
Challenges	Difficult to fully automate; e.g., disconnecting wires, unscrewing bolts. Difference in EV battery design can limit transferability of robot/cobot parameters between disassembling different EV batteries.
Current Status	Pilot projects, pilot disassembly lines, limited throughput.
Key Player / Project Examples	BATRAW; LIEBHERR; ELECTRIFYINGFUTURE BeePlanet factory; CIRCU LI-ION; HUAYOU

EV Battery Trends and Impacts on Second-life EV Batteries

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Trend	Impact on second-life EV batteries
1. Removal / absence of battery module (cell-to-pack or cell-to-chassis design).	Could reduce cell-level disassembly times/costs. However, more likely to be a hindrance due to use of more structural adhesives, welds, etc., which necessitate the use of solvents or heat to remove.

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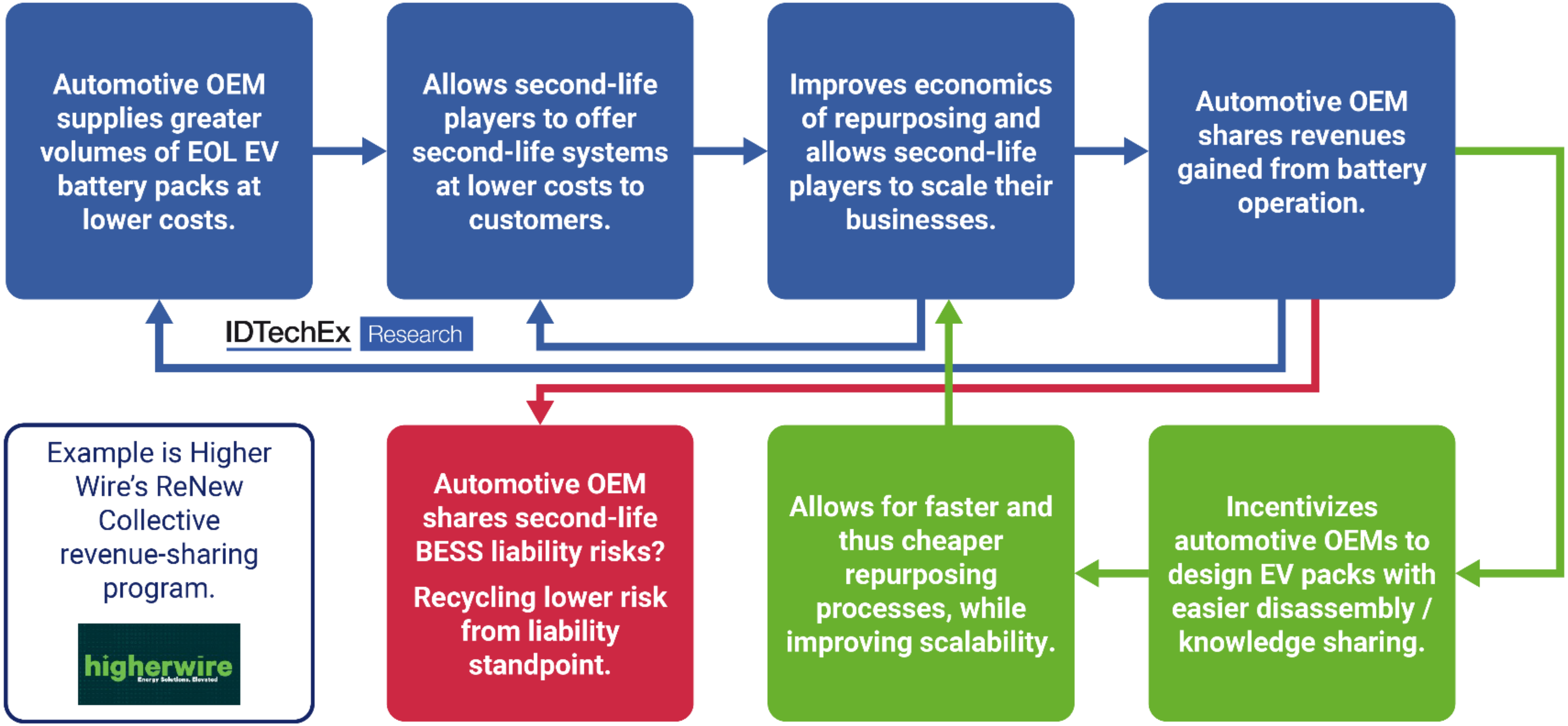
+ Other key battery trends, which are discussed and their impact on second-life EV batteries or repurposing analyzed in IDTechEx's Second-life Electric Vehicle Batteries 2025 – 2035 market report.		

EV Battery Design for Circularity and Revenue Sharing Models: A Hope or Future Reality?



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Access to 30+ Company Profiles: Repurposers and Second-life Battery Diagnosticians / Assessment Players

This list includes second-life EV battery repurposers, battery performance testers and modelers for second-life applications, and several Li-ion battery recyclers that are already participating, or may look to participate, in the second-life EV battery market.

1. Allye Energy
2. B2U Storage Solutions (2023)
3. B2U Storage Solutions (2022)
4. BBB Industries / TERREPOWER
5. BeePlanet Factory (2024)
6. BeePlanet Factory (2022)
7. Betteries (2024)
8. Betteries (2022)
9. Cidetec
10. Circu Li-ion
11. Circunomics
12. Connected Energy (2024)
13. Connected Energy (2023)
14. Covalion
15. Currents
16. Eatron Technologies
17. ECO STOR AS (2024)
18. ECO STOR AS (2023)
19. Ecobat
20. Exitcom Recycling
21. Higher Wire
22. Huayou Recycling
23. Liebherr-Verzahntechnik
24. Oorja Energy
25. Reefilla
26. ReJoule (2024)
27. ReJoule (2023)
28. Relectrify
29. RePurpose Energy
30. Safion
31. SK tes
32. Smartville (2024)
33. Smartville (2023)
34. Spiers New Technologies
35. volytica diagnostics
36. Zenobē

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Summary and IDTechEx Outlook

As the availability of retired EV batteries will grow over the coming decade, IDTechEx forecasts the second-life EV battery market will be valued at US$4.2B by 2035. Market growth has remained steady, though key trends could help to reduce costs of repurposing and improve how competitive these technologies are against first-life Li-ion BESS.

Drivers

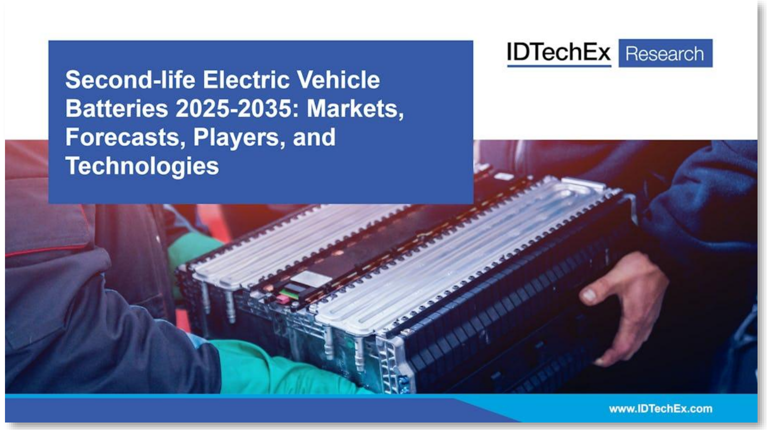
- Repurposing looks to maximize the value of the EV battery and the raw materials initially used, extend its lifetime, and delay recycling.
- Could also help to reduce the battery associated CO_2 (raw materials, transport, recycling) per kWh of total energy dispatched by the battery over the course of its lifetime.

Opportunities

- Array of opportunities to reduce repurposing costs – B2B marketplaces for lower retired EV battery costs, faster grading technologies, automated battery disassembly, and more.
- Market will grow more significantly as availability of retired EV batteries increases.
- Potential for revenue sharing opportunities and for repurposers to capitalize on using other business models (BSaaS, renting).

Challenges

- Reducing price of first-life Li-ion BESS has made it increasingly difficult for second-life repurposers to offer their technologies at competitive prices.
- Lack of policies or regulations that incentivize or encourage repurposing, with more of a focus in key regions being on recycling and recovering critical raw materials.
- Lack of standardized EV battery design and cell-to-pack designs can complicate repurposing processes and add to costs.



Second-life Electric Vehicle Batteries 2025 – 2035: Markets, Forecasts, Players, and Technologies

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